

Checking and Revising Reasoning from Uncertain Evidence

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Program Officer: Dr. Ben Robinson, AFOSR/RESC

Principal Investigator: Xinya Du

Assistant Professor, Department of Computer Science

The University of Texas at Dallas

Email: xinya.du@utdallas.edu

Telephone: +1 (972) 883-2634

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1. Research Problem and Central Idea

A reasoning system can apply every logical rule correctly and still reach an unsupported conclusion because it misread the evidence. Consider a maintenance report: a sensor failed at 10:02, and a power outage was reported at 10:05. Treating the report time as the outage time would incorrectly rule out the outage as a possible cause. If a later correction places the outage at 9:58, that objection disappears, but timing alone still does not establish causation. The system must distinguish what the report says, what it implies, and what remains unknown.

We ask: *When a system may have misread the evidence, what should it check before accepting an answer, and what should it reconsider after a correction?* The correct interpretation may be absent from the model’s candidates. A single misread passage may supply several incorrect premises. Reusing a check or revising only one argument can therefore save computation while spreading an error.

Our hypothesis is that a system can check and revise its conclusions more reliably if it records both the sources of its premises and the logical links between them. We will test whether this reduces unsupported answers at matched computational cost and answer coverage—the fraction of questions answered. The three aims use this record to keep track of possible interpretations, choose what to check, and revise conclusions.

Foundation and contribution. Our prior work on structured reasoning and temporal relation extraction provides methods for representing evidence [4, 6]. MEQA provides questions about events and their explanations [7]; FG-PRM identifies errors within mathematical reasoning steps [8]. Search Wisely studies when additional retrieval is needed [12]. Here, we will extend these methods to checking and revising interpretations of evidence. Solver-assisted reasoning, truth maintenance, and selective generation provide established foundations [1, 5, 9, 10]. We will study how reliably and efficiently these methods can work when their premises are incomplete or affected by the same reading error.

2. Research Aims and Technical Approach

Aim 1: Represent what the evidence does and does not establish

Each argument will record source passages, their candidate interpretations, explicit assumptions, and the rules connecting premises to conclusions. We will begin with finite arguments about entities, event order, and fixed conditional rules. Uncertain entity matches and event times will be premises to check, rather than unquestioned inference rules. In the example, the representation will distinguish the time an outage occurred from the time it was reported.

A language model will propose a limited set of joint interpretations, each giving a consistent reading of the relevant passages. Separate checks will assess inferences and textual support. Rule checks will be exact within the specified formal language; evidence checks will be learned and empirically validated. A rule chosen on development data will determine which interpretations to keep. The system will accept an

answer only if a checked argument supports it under every interpretation it keeps. Every evidence check required by those arguments must also be resolved. If no interpretation remains, the answers disagree, or one reading lacks a valid argument, the system must check further or withhold an answer. Possibilities that depend on unverified assumptions will be labeled as such.

This criterion has a limited but useful guarantee: with sound rules, an accepted answer follows under the correct joint reading if that reading remains in the candidate set. Rule soundness alone does not establish that the evidence was understood. We will study how generating and filtering interpretations affects cost and the risk of losing the correct reading, including cases where several premises fail together. Controlled examples and human annotations will measure these omissions independently of the checker. The system will derive its answer and explanation from the checked arguments. These show the support for the answer; they do not describe the language model’s internal computations.

Aim 2: Spend checking effort where it can change the decision

In the example, clarifying whether 10:05 is a report time may resolve several premises at once. Checking each premise separately may waste effort, while accepting them together may repeat a mistake. We will compare checking one premise at a time with examining a shared passage and its possible readings together.

Cost-sensitive testing and model-based diagnosis already study how to choose informative tests [2, 3]. Work on sequential anomaly search also accounts for the cost of switching from one source to another [11]. We will account for measured costs of checking and loading source context when deciding whether to continue examining a passage or move to another. Our problem also requires logical support for an answer, and the correct interpretation may be missing. The guarantees for anomaly search therefore do not transfer automatically.

We will begin with small acyclic argument graphs and finitely many checks. Each evidence check returns supported, contradicted, or unresolved. Policies will choose the next check using its cost, its estimated effect on the acceptance decision, and the premises that share its source. Learned scores will prioritize checks, not certify their outcomes. The goal is to answer as many questions as possible within a checking budget while keeping the error rate among accepted answers below a specified level. An unresolved result will identify the premise or missing check that prevents acceptance.

For the formal analysis, we will first specify how check outcomes depend on one another, allowing premises from the same source to have correlated errors. Dynamic programming over finite checking histories will provide exact reference policies for small cases. We will examine how the number of retained readings and the size of shared-source groups affect this computation, then compare less costly policies with these exact solutions. A key question is when grouping checks saves cost and when it hides a decisive alternative. We will change the assumed outcome model and remove correct interpretations to test how

the policies fail when their assumptions are wrong. Document experiments will test the limits of these analyses.

Reliability will be assessed for the complete acceptance procedure. After fixing the policy on development data, independent validation cases will support finite-sample bounds on the unsupported fraction of accepted answers. This uses established statistical tools; correlations inside an argument are not treated as independent validation samples. Tests on new document sources will show whether the results extend beyond the validation data. We will report coverage and uncertainty intervals.

Aim 3: Reopen the right evidence when an interpretation changes

Classical truth maintenance records which conclusions depend on which assumptions [1]. Correcting a source can also change premises that have no direct logical link. Discovering that a report uses reporting times rather than event times may require revisiting several timestamps, including ones in otherwise unrelated arguments. We will therefore record both logical links and premises whose interpretation depends on the same context.

The central experiment will compare three update procedures: revise only conclusions that logically depend on the corrected premise; also recheck premises that share its source context; or recompute every interpretation and argument. After a correction, the system will recheck the affected source group before updating conclusions that depend on it. Conclusions with independent valid support can be preserved. In the outage example, removing the temporal objection leaves causation unresolved unless other evidence supports it. Every proposed repair must again pass the acceptance procedure; invented premises or weakened rules will not count as repairs.

We will analyze sufficient conditions for local revision to agree with full recomputation under a fixed interpretation procedure, including completeness of the recorded context dependencies. Equivalence for fixed premises is an established baseline. We ask how much evidence needs to be reinterpreted when those links are uncertain. We will compare passage-level grouping with wider document-level rechecking, measure failures caused by missing links, and study when broader rechecking costs as much as starting over. Checker-guided refinement, including solver feedback as in Logic-LM [10], will be the initial repair mechanism. We will consider training a model to propose repairs if feedback on specific errors proves useful. Explicit limits on checks and repair attempts ensure termination; reaching a limit returns unresolved, not successful repair.

3. Evaluation and Measures of Success

Controlled problems will have known facts and valid derivations, with variation in reasoning depth, alternative support, shared-source errors, missing interpretations, and evidence corrections. Independently checked text versions will separate reading errors from inference errors. We will extend MEQA with human judgments of support for each step and cases where the evidence is corrected. Ground truth will

come from the controlled task specifications and independent human review, never the system’s own checkers. Held-out rule combinations and document sources will test generalization. Questions outside the rule library will count as unresolved rather than being discarded.

Baselines will include direct language-model reasoning, FG-PRM step scoring, solver-assisted reasoning with checker feedback, selective generation, and truth maintenance with uniform or exhaustive evidence checking. All systems will receive the same evidence and be asked to report their supporting steps in the same format. An answer with missing or unclear support will not count as verified. We will match the base model and total computation, including generation, checking, and revision; compare error at matched coverage; and report reliability-cost curves. Ablations will separate logical dependencies, shared-source grouping, and adaptive check selection.

Our main measure is the fraction of accepted answers supported by an invalid inference or a premise that the evidence does not justify. We will also report this fraction among *correct* accepted answers, using cases where an invalid step can accidentally yield the right answer. Other measures are answer accuracy and coverage, interpretation accuracy, omission of correct readings, and repair success and cost. We will change necessary evidence and test whether conclusions change accordingly, scoring cases with other valid support separately. Success requires fewer unsupported answers at comparable coverage and cost, and cheaper correct updates than full recomputation. We will report cases where shared-source checking adds cost without improving reliability.

4. Potential Impact, Team, and Development Plan

Potential impact on DAF and DoW capabilities. Maintenance and anomaly investigations require combining reports that may be incomplete, ambiguous, or later corrected. These methods could help analysts distinguish supported explanations from possibilities, locate supporting reports and assumptions, and identify the next check needed. When a timestamp or equipment identifier is corrected, the system would show which conclusions changed and why, with links to the evidence. This basic research would establish when such updates can safely reuse earlier work and when broader review is necessary. It directly addresses the Computational Cognition and Machine Intelligence portfolio’s interests in interpretable foundations, uncertainty, verification, and repair of faulty arguments.

PI Xinya Du will lead the work at UT Dallas with graduate and undergraduate research students. The estimated three-year budget is \$450,000, including institutional indirect costs, for personnel, computation, annotation, and dissemination. No external collaborators or subawards are planned.

Year 1 will build a small end-to-end system covering all three aims, define the rules and how interpretations are represented, and test it on controlled ambiguities and corrections. **Year 2** will develop check-selection policies, analyze restricted cases, and evaluate shared-source errors on annotated documents. **Year 3** will refine revision policies and test reliability and cost across new sources. Deliverables include reproducible code, checker specifications, annotated data, publications, and presentations.

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